ADS MANAGEMENT SYSTEM

Project Overview

Read this first before the Developer Handoff Document

Item	Detail
Project name	Ads Management System
Scope	Part 3: Analytics & Monitoring + Part 4: Conversion Tracking
Number of brands	10 or more — all built on the same tech stack
Primary goal	Make Meta ads optimize for real paid sales — not just messages
Secondary goal	Catch ad performance problems early before they become revenue problems
Automation engine	n8n self-hosted on Hetzner VPS
Before you build	Run all 32 POC tests first — some document sections may change based on results
Full technical details	See: Ads Management System Master Developer Handoff Document

1. The problem this system solves

Problem 1 — Meta is optimizing for the wrong thing

This business sells high-ticket products — chandeliers, lighting, furniture. The typical customer journey looks like this:

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Customer sees a Meta ad
    ↓
Customer clicks the ad and visits the Shopify store
    ↓
Customer messages via WhatsApp, Viber, Messenger, or Email
    ↓
Sales team sends a quotation
    ↓
Customer pays — days or weeks later — through Zoho Books
    ↓
Meta never finds out the sale happened
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Meta's algorithm only sees the click and the message. It has no idea whether that customer eventually paid ₱500 or ₱500,000. So Meta optimizes for cheap messages — not for real buyers.

The result: ad budget flows toward ads that generate lots of messages from people who never buy, while ads that attract genuine high-value buyers may be starved of budget because their revenue is invisible to Meta.

Problem 2 — No early warning system

When ad creatives start to fatigue, CTR declines gradually before collapsing. Without monitoring, nobody notices until Zendesk inquiries have already dropped significantly — by which point significant budget has already been wasted.

This business experienced exactly this: Zendesk inquiries dropped from 300-400 per month to 80-100 per month before the problem was identified. The root cause was Meta ad creative fatigue that went undetected for weeks.

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WITHOUT a monitoring system: the team only discovers the problem when inquiries collapse.
WITH a monitoring system: an email alert fires when CTR first starts declining — weeks earlier.
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2. The solution — 2 systems working together

This project builds two systems that run in parallel and solve both problems simultaneously.

System 1 — Analytics & Monitoring (Part 3)

An automated monitoring system that watches Meta ad performance daily and sends email alerts when metrics fall below defined thresholds. It catches creative fatigue at the earliest signal — before it hits traffic, before it hits inquiries, before it hits revenue.

What it monitors	How it monitors	What happens when threshold is breached
CTR — Click-Through Rate	Compares this week's CTR vs last week's CTR on every active ad	Email alert fired — monitoring person and brand team notified
CPC — Cost Per Click	Compares this week's CPC vs the ad's own 30-day average	Email alert fired — review ad spend efficiency
Frequency — how often same person sees the ad	Monitors adset-level frequency over 7-day window	Email alert fired — audience is burning out
Impressions vs Spend	Checks if impressions drop while spend stays flat	Email alert fired — possible Meta delivery problem
Inquiry volume	Tracks daily Zendesk ticket count vs GA4 sessions	Dashboard view — shows if inquiries are declining relative to traffic

System 2 — Conversion Tracking (Part 4)

A complete tracking pipeline that follows every customer from the moment they click a Meta ad all the way to the moment they pay — and sends that revenue data back to Meta so the algorithm learns which ads generate real sales.

Ad click → Shopify captures the click ID → customer clicks WhatsApp/Viber/Messenger/Email
→ redirect page logs tracking data to n8n → Zendesk ticket gets enriched with tracking data
→ customer requests quote → Zoho Books estimate auto-created with tracking data
→ customer pays → staff creates Sales Order → n8n sends ₱[amount] to Meta CAPI
→ Meta learns which ads make real money → budget flows to better ads automatically

The end result: Meta stops optimizing for cheap messages and starts optimizing for real paid revenue. Ad spend becomes more efficient. Scaling becomes more reliable.

3. How it works — step by step

Step 1 — Customer clicks a Meta ad

Meta assigns a unique tracking ID called an fbclid to every ad click. When the customer lands on the Shopify store, a small invisible script captures this fbclid and generates a session token — a unique ID for this visit, like HC-20260714-18291. Both are saved silently in the customer's browser.

Step 2 — Customer clicks a contact button

Instead of linking directly to WhatsApp, the contact button passes through an invisible redirect page first. In under 200 milliseconds, the redirect page sends the session token and fbclid to n8n (the automation engine). Then WhatsApp opens normally. The customer notices nothing.

Step 3 — Customer sends a message

The message creates a ticket in Zendesk. n8n immediately finds that ticket and writes the session token, fbclid, channel, and brand into the ticket's custom fields. The inquiry is now fully linked to the original Meta ad click.

Step 4 — Customer requests a quotation

When the customer uses words like 'quote', 'how much', or 'magkano', Zendesk automatically detects it. n8n responds by creating a blank estimate in Zoho Books — already pre-filled with all the tracking data — and emailing the sales team. The sales agent just adds products and prices.

Step 5 — Customer pays

The customer pays. Staff converts the quotation to a Sales Order in Zoho Books. The moment that happens, Zoho Books sends an automatic notification to n8n.

Step 6 — Meta gets the revenue data

n8n reads the tracking data from the Sales Order, hashes the customer's email and phone for privacy, and sends a Purchase event to Meta CAPI: this ad click generated a sale worth ₱[amount]. Meta now knows which specific ad, audience, and creative led to that real paid sale.

Step 7 — Meta gets smarter over time

As more Sales Orders are tracked and fed back to Meta, the algorithm builds a picture of what a real buyer looks like. It starts showing ads to more people who match that profile — and less to people who message but never buy. Ad performance improves. Cost per real sale decreases.

4. The platforms and what each one does

Platform	Role in the system	Who sets it up
Meta Ads	Generates ad clicks and fbclid tracking IDs. Receives revenue data via CAPI and optimizes accordingly.	Already active — developer configures CAPI Dataset
Shopify (OS 2.0)	Captures fbclid and session token when customer arrives. Hosts the redirect pages for contact buttons. Must be upgraded to OS 2.0 before development starts.	Developer adds tracking script and redirect pages
Zendesk (shared)	Central inbox for all customer inquiries across all brands. Stores all tracking data in custom fields. Detects RFQ keywords automatically. Shared across all brands — brand tagging separates tickets.	Developer creates custom fields, triggers, and brand tagging
Zoho Books	Creates estimates and Sales Orders. Custom tracking fields carry the session token and fbclid from estimate through to Sales Order. Sales Order creation triggers the CAPI event.	Developer creates custom fields and webhook
n8n (Hetzner VPS)	The automation engine. Connects everything. Receives webhooks, enriches tickets, creates estimates, sends CAPI events, runs daily audit. Self-hosted on a Hetzner cloud server.	Developer installs server and builds all 11 workflows
Zoho Analytics	Monitoring dashboard. Pulls data daily from Meta Ads, Shopify, GA4, Zendesk, and Zoho Books. Fires email alerts when thresholds are breached.	Developer connects all 5 data sources and builds dashboards
Google Analytics 4	Traffic and session data. The only way to see how many sessions came from Meta paid ads vs other sources. Must be installed on every brand Shopify store.	Developer installs GA4 on each store

5. Multi-brand architecture

This system is built for 10 or more brands from day one. All brands share the same tech stack, the same target market, and the same contact channels setup — which means the system is built once for the primary brand, then replicated for each additional brand.

How the brands are structured

PRIMARY BRAND workspace (Zoho Analytics) → feeds into → MASTER DASHBOARD
BRAND B workspace (Zoho Analytics) → feeds into → MASTER DASHBOARD
BRAND C workspace (Zoho Analytics) → feeds into → MASTER DASHBOARD
BRAND D workspace (Zoho Analytics) → feeds into → MASTER DASHBOARD
... and so on for all brands

MASTER DASHBOARD = what the monitoring person watches every day
BRAND WORKSPACES = what each brand's marketing team watches

Who does what

Role	Responsibility	Does NOT do
Monitoring person (1 person)	Watches Master Dashboard daily. Receives all alerts across all brands. Triages severity. Escalates to correct brand team.	Does not execute fixes — only identifies and escalates
Brand marketing team (per brand)	Receives escalated alerts for their brand. Executes the fix — pauses ad, rotates creative, expands audience.	Does not monitor other brands
Business owner	Receives Critical alerts only. Involved when multiple brands are simultaneously in critical state.	Not involved in day-to-day monitoring

Alert routing

Alert level	When it fires	Who receives it
WATCH	CTR dropped 20-29% vs prior week on same ad. Frequency above median (2.37).	Monitoring person only
WARNING	CTR dropped 30-44%. CTR below 0.67% absolute. CPC up 30-49%. Frequency above 3.5.	Monitoring person + relevant brand team
CRITICAL	CTR dropped 45%+. CTR below 0.43% absolute. CPC up 50%+. Frequency above 4.5. Delivery problem.	Monitoring person + brand team + business owner

6. Build sequence

Build in this exact order. Do not skip steps or build out of order. Each step depends on the one before it.

PHASE 1 — BEFORE ANY CODE IS WRITTEN (1-2 weeks)
- Step 1: Upgrade all brand Shopify stores to Online Store 2.0
- Step 2: Run all 32 POC tests — verify every assumption before building
- Step 3: Fix any test failures — update the developer handoff if architecture changes

PHASE 2 — PRIMARY BRAND BUILD (2-3 weeks)
- Step 4: Set up Hetzner VPS + install n8n with Docker and Traefik SSL
- Step 5: Install GA4 on primary brand Shopify store + add UTMs to all Meta ads
- Step 6: Connect all 5 Zoho Analytics connectors for primary brand
- Step 7: Set up Zendesk — custom fields, brand tagging triggers, RFQ detection
- Step 8: Add Shopify tracking script and redirect pages to primary brand store
- Step 9: Build all 5 n8n workflows
- Step 10: Set up Zoho Books — custom fields and Sales Order webhook
- Step 11: Configure Meta CAPI Dataset and test events
- Step 12: Build Zoho Analytics dashboards and configure all 11 email alerts
- Step 13: Run full end-to-end QA test — all 7 test cases must pass

PHASE 3 — MULTI-BRAND ROLLOUT (1-2 days per brand)
- Step 14: Replicate for each additional brand — one brand at a time
- Step 15: Build Master Dashboard in Zoho Analytics
- Step 16: Run 30-item final acceptance checklist

7. Complete tech stack

Platform	Technology	Cost	Notes
Automation server	Hetzner VPS CX22	~€4/month	Shared across all brands. Runs n8n 24/7.
Automation engine	n8n (self-hosted)	Free (self-hosted)	11 workflows. Connects all platforms.
SSL/Proxy	Traefik (via Docker)	Free	Built into n8n official Docker setup.
BI & Monitoring	Zoho Analytics	Existing subscription	One workspace per brand + Master workspace.
CRM	Zendesk	Existing subscription	Shared across all brands. Brand tagging separates tickets.
Accounting	Zoho Books	Existing subscription	One organization per brand.
Ecommerce	Shopify (OS 2.0)	Existing subscription	Must upgrade to OS 2.0 before dev starts.
Traffic data	Google Analytics 4	Free	One property per brand store.
Ad platform	Meta Ads	Ad spend only	One CAPI Dataset per brand.

8. Documents in this project

Three documents have been prepared for this project. Read them in this order:

1. This document — Project Overview. Read first. Understand what is being built and why.
2. POC Test Plan (32 tests). Run all 32 tests before development starts. Fix all blockers before proceeding.
3. Master Developer Handoff Document. The complete technical specification. Read after the POC tests are done and any architecture changes from failed tests have been incorporated.

DO NOT start development from the Master Developer Handoff Document alone.
Run the 32 POC tests first. Some sections of the handoff may change based on test results.
The handoff document is the correct architecture based on verified assumptions —
but the POC tests will confirm which assumptions hold in your actual live accounts.

9. Important rules for the developer

Before writing any code

- Upgrade all Shopify stores to OS 2.0. The redirect page system requires OS 2.0 JSON templates. Any tracking code written on OS 1.0 will need to be rewritten after the upgrade.
- Run all 32 POC tests. Do not skip any blocker tests. If a blocker fails, fix it before proceeding.
- Read both the POC Test Plan and the Master Developer Handoff Document in full before starting.

While building

- Build for the primary brand first. Verify it completely. Then replicate for other brands.
- Never hardcode credentials, API keys, or tokens into workflow code. Store all credentials in n8n's credential store.
- The FR Stores Reach campaign must be excluded from all Zoho Analytics alert queries. It uses a Reach objective — its naturally low CTR will cause constant false alerts if not excluded.
- Every Meta ad destination URL must have UTM parameters before the campaign goes live. Without UTMs, GA4 cannot identify Meta paid traffic.
- Never overwrite an existing session token on a Zendesk ticket. If a token already exists, preserve it.

Staff dependencies

This system has operational dependencies that the developer cannot fix. Before go-live, the business owner must ensure:

- Staff capture customer email and phone in Zoho Books on at least 80% of Sales Orders
- Staff create Sales Orders within 24 hours of receiving payment
- All staff understand never to fill in tracking fields for customers who did not arrive via a Meta ad
- Every brand has its own distinct WhatsApp number, Messenger page, Viber account, and inquiry email

The system is only as good as the data flowing through it.

A technically perfect system fed by incomplete staff data will underperform.

These operational requirements must be in place before go-live.